

Capstone Project: Banking Risk & Credit Portfolio Intelligence Platform

Business Scenario: Building a Secure Data Pipeline with **AWS, Snowflake, Informatica Intelligent Cloud Services (IICS), and Tableau** for a unified **Banking Risk & Credit Portfolio Intelligence Platform** for Aurora National Bank.

Project Overview: General

The goal of this capstone project is to design, develop, and deploy a secure and automated data pipeline that extracts, transforms, and loads (ETL) loan and collections data from AWS S3 into Snowflake using **Snowpipe** and **Snowpark (Python)**. The project also includes securing sensitive data using **Data Masking**, setting up **AWS SQS** for event notifications, and leveraging **Informatica IICS** for orchestrated data integration. **Tableau** will be used to create analytical dashboards for credit portfolio risk, delinquency, and collection performance.

As a Snowflake Developer in a large retail and SME banking organization, your role is to build and visualize an enterprise-grade Credit Risk Platform that consolidates origination, bureau, and collection data into a single analytical layer for the **Risk Analytics Division**.

Tools Utilized:

- **Snowflake:** For data loading/unloading, Storage Integration using Snowpipe, and automation of tasks and streams.
- **IICS:** For creating Mapping and Taskflows to ingest and transform banking datasets into Snowflake.
- **Tableau:** For developing dashboards that deliver insights into loan performance, delinquencies, and recoveries.

Objective:

- Design and implement a robust, end-to-end data integration pipeline using **Snowflake, IICS, and Tableau**.
- Develop dashboards and reports that visualize **loan lifecycle trends, delinquency buckets, and recovery metrics**.
- Ensure scalability, availability, version control, and security for all data operations.
- Optimize Tableau extracts and Snowflake queries for better performance and resource utilization.
- Document the complete architecture and workflow for organizational reusability and audits.

Assumptions:

- The project team consists of 4–5 associates skilled in Snowflake, IICS, and Tableau.
- Access to AWS, IICS, and Tableau environments is provided.
- Team members are assumed to have intermediate-to-advanced knowledge of banking datasets and ETL design.

Expectations:

- All implemented workflows must be **error-free** and **validated** with test reports.
- Include **data validations** and **exception handling** wherever applicable.
- Follow **Agile practices** with sprints, scrums, and burn-down tracking.
- Incorporate non-functional parameters like **Security, Scalability, and Availability**.
- Carefully design normalized data models and present entity relationships.
- Ensure all Snowflake and Tableau objects (pipes, tasks, dashboards) are robust against pipeline failures.
- Identify and present the **Unique Selling Proposition (USP)** — for example, *“Automated Early-Delinquency Alert System.”*
- Complete within **4 working days**, with **2 additional days** for presentation and refinements.
- Document all configurations, workflows, and results comprehensively.
- Maintain collaboration and regular status updates with stakeholders.
- Adhere strictly to best practices for integration, visualization, and optimization.

Use Cases:

1. **Data Integration:** Design ETL pipelines in IICS and Snowflake to integrate **loan, customer, and collections** data.
2. **Risk Visualization:** Build Tableau dashboards showing delinquency movement, recovery trends, and regional portfolio mix.
3. **Scalability and Reliability:** Ensure automated incremental updates using Snowpipe and Streams.
4. **Optimization:** Improve Tableau performance through extract tuning and materialized views.
5. **Security:** Apply masking policies for PII and restrict portfolio data by business region.

Submission Guidelines:

- Provide regular updates and track progress using Agile boards.
- Submit documentation for dashboards, ETL workflows, and data design.
- Present key findings and business recommendations to stakeholders.
- Demonstrate working pipelines and interactive Tableau dashboards.
- Participate in collaborative review and feedback sessions.

Banking Risk & Credit Portfolio Intelligence Platform: Technical Specification

Case Study Context:

You are part of the **Risk Analytics Division** at **Aurora National Bank**, operating across multiple countries with diversified retail and SME loan portfolios. Leadership requires a centralized **Credit Risk & Portfolio Intelligence Platform** to track loan origination quality, monitor delinquency patterns, and optimize collection strategies.

You are tasked with building this platform using:

- **Snowflake** for cloud data warehousing and governance.
- **IICS** and **Snowpipe** for hybrid ingestion and automation.
- **Tableau** for executive dashboards and self-service analytics.

Business Objectives

- Consolidate origination, bureau, collateral, and collection datasets into a unified risk warehouse.
- Build automated, incremental pipelines with validation and alerts.
- Implement data masking and RLS for regulatory compliance.
- Deliver real-time dashboards for credit risk leadership to monitor KPIs.

Input Datasets (All Provided as .csv):

Dataset Name	Description	Source
Loan_Accounts.csv	Loan contracts, origination, and repayment terms	Manual Upload
Customer_Master.csv	Customer demographics and identifiers (masked PII)	Static Table
Bureau_Scores.csv	External credit bureau scores and ratings	S3 Bucket
Collections_Events.csv	Collection activities, DPD updates, payments, PTPs	Streaming to S3
Collateral_Register.csv	Details of pledged collaterals and valuations	Manual Upload
Region_Hierarchy.csv	Country, Region, Branch mapping	CSV Upload

Project Phases and Detailed Tasks

PHASE 1: 🔄 Data Acquisition and Loading

Task 1: Setup Snowflake Environment

- Create warehouse, database, schema.
- Setup roles, users, and apply least-privilege access.
- Configure resource monitors to control compute cost.

Task 2: Load Flat Files via SnowSQL

- Use PUT and COPY INTO commands for:
 - Customer_Master.csv
 - Region_Hierarchy.csv
 - Collateral_Register.csv
- Define staging tables with data-type mapping and validation rules.

Task 3: Core Data Ingestion via IICS

- Extract data from **Loan_Accounts** (core system).
- Transform and load into **stg_loan_accounts** in Snowflake.
- Schedule refresh using **IICS Taskflow** (daily).

Task 4: Real-Time Collection Feed via Snowpipe

- Create **External Stage** linked to S3 bucket.
- Define **File Format (CSV)** with auto-date detection.
- Build and enable **Pipe** for **Collections_Events.csv** ingestion.
- Validate ingestion by simulating new file drops into S3.

PHASE 2: 🏗️ Data Modeling and Star Schema

Task 5: Build Dimension Tables

- **DimCustomer:** CustomerID, Name (masked), Segment, RiskBand, Region.
- **DimProduct:** ProductID, LoanType, Tenor, APRTier.
- **DimRegion:** Country, Zone, Branch, RegionHierarchy.
- **DimCollateral:** CollateralID, CollateralType, ValuationBand.

Task 6: Build Fact Tables

- **FactLoan:** LoanID, OriginationAmount, APR, Tenor, CurrentBalance, joins dimensions.
- **FactCollections:** DPD, ContactResult, PTP, Cure%, joins with customer and region dimensions.
- **FactBureau:** BureauScore, Rating, Month, joins with DimCustomer.

Task 7: Apply Surrogate Keys and Constraints

- Add surrogate keys to all dimension tables.
- Apply foreign key constraints and use **MERGE** for incremental upserts.

PHASE 3: ⚙️ Incremental Automation Using Streams, Tasks, and MVs

Task 8: Stream Setup

- Create **STREAM** on stg_loan_accounts and stg_collections_events.
- Track changes with METADATA\$ACTION to detect inserts/updates.

Task 9: Task Automation

- Define **hourly TASK chains** to:
 - Insert new loans into FactLoan.
 - Update latest DPD and recovery data in FactCollections.
 - Record pipeline metadata in audit_log.

Task 10: Materialized Views for Performance

- mv_vintage_performance: Aggregates loss ratio by origination month.
- mv_roll_rate: Tracks transitions between delinquency buckets.
- mv_collections_effectiveness: Summarizes PTP kept %, cure rates.

PHASE 4: 🛡️ Security, Governance and Sharing

Task 11: Data Masking and RLS

- Apply **MASKING POLICY** on:
 - DimCustomer.customer_name
 - DimCustomer.email
 - DimCustomer.phone
- Implement **ROW ACCESS POLICY** restricting portfolio access by Region or Country.

Task 12: Secure Data Sharing

- Create **SECURE SHARE** exposing:
 - mv_vintage_performance
 - mv_collections_effectiveness
- Test accessibility via **reader account**.
- Document sharing process and validation.

PHASE 5: 📊 Visualization Using Tableau

Task 13: Data Connection

- Connect Tableau to Snowflake using **live/extract** mode.
- Build calculated fields for DPD buckets, cure flags, and loss ratios.

Task 14: Design Dashboards

Dashboard	Visuals & Features
Portfolio Overview	KPIs (NPL%, Write-off%, Recovery%), trend charts.
Vintage Trends	Line/Area charts for losses by origination month.
Collections Command Center	Agent performance, PTP%, Cure%, heatmaps.
Regional Risk Heatmap	PAR%, NPL%, Recovery% by geography.
Loss & Recovery	Waterfall and ratio visuals for net loss tracking.

Task 15: Interactivity Features

- Region/Product filters.
- Drilldowns (Country → Branch → Agent).
- Alert indicators for high-risk thresholds.
- Parameter toggles for time-window adjustments.

PHASE 6: 📄 Documentation and Submission

Task 16: Reporting

- Create architecture diagrams (Snowflake + IICS + Tableau).
- Provide ER Diagram and pipeline workflow.
- Include proof of security policies, masking, and shares.
- Submit SnowSQL scripts, Tableau workbooks, and CSVs.

Deliverables

Type	Description
SQL Scripts	DDL, Stream, Task, Policy scripts.
Data Files	Clean .csv source datasets.
Tableau Dashboards	.twbx or PDF exports.
Documentation	Word/PDF project report.
Screenshot Proofs	Security, sharing, and pipeline validation.